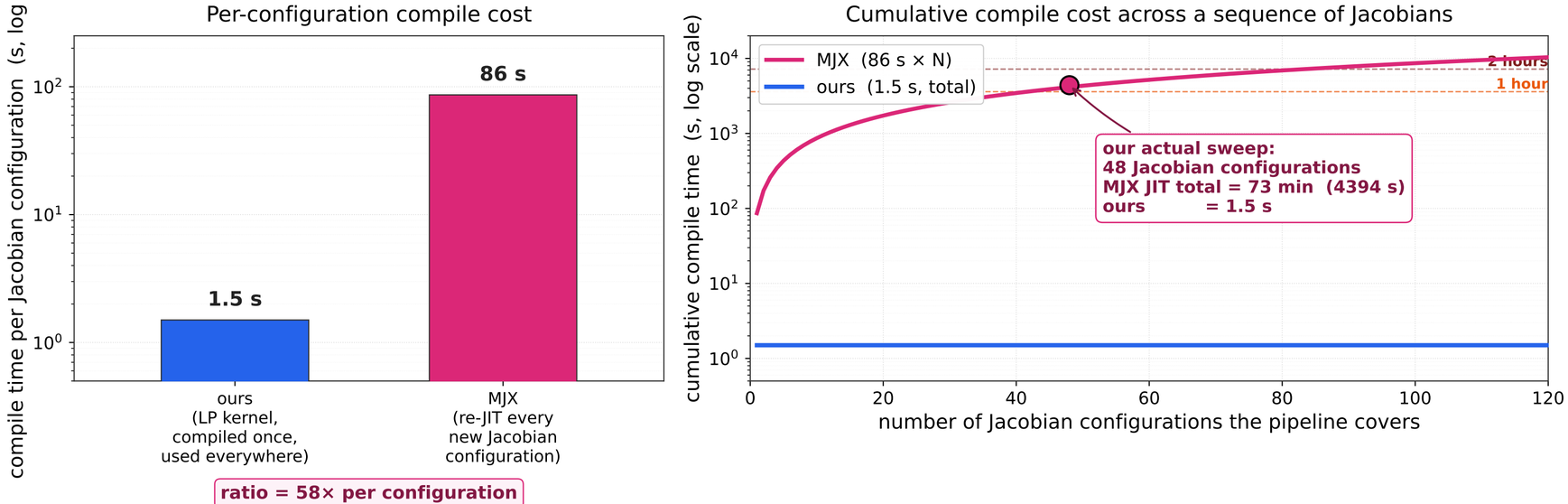
Compile time scales with the # of Jacobians your pipeline covers · ours = 1.5 s once · MJX = 73 min for our 48-Jacobian sweep



Each Jacobian configuration is a distinct (robot × precision × batch size × problem-mode) tuple — i.e. one new XLA trace for MJX. ours = data-driven kernel, compiled once at session start, reused across every configuration with no recompile. Empirical median MJX JIT-per-configuration across our 48-config sweep = 86 s.